

## PULSE RADIOLYSIS STUDY OF TRANSIENT SPECIES IN LiCl-KCl MELT

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(Received 29 September 1985)

**Abstract**—Pulse radiolysis study on LiCl-KCl melt was carried out. Results showed that two kinds of transient species were formed in this system. From the analysis of their nature, one was attributed to the solvated electron  $e_s^-$ , and the other was attributed to the molecular radical ion  $Cl_2^-$ . The properties of the reactions resulting from these species were discussed.

### INTRODUCTION

The radiolytic study of some alkali halide melts was reported by Pikaev *et al.*<sup>(1)</sup> They observed transient species in an irradiated system using the microsecond pulse radiolysis technique, and discussed their nature.

In the present study, we attempted to observe the transient species in LiCl-KCl melt by the nanosecond pulse radiolysis technique. As the LiCl-KCl melt is considered to be a highly useful material, for example, as a coolant for the molten salt reactor, it is important to discuss the stability of this material for irradiation.

### EXPERIMENTAL

For the pulse radiolysis experiments, the electron pulse from the linac installed at Hokkaido University (producing 11 ns pulse of 45 MeV electrons) was used. The dose per pulse was 50–85 Gy, which was measured by the KCNS method. Samples were put into a quartz glass ampoule and sealed after evacuating them for a long time at high temperature up to their melting points.<sup>(2)</sup> Experiments were carried out by placing the ampoule in an electric heater, which was placed at the exit of the linac. Temperature of the samples was measured by chromel–alumel thermocouple.

### RESULTS AND DISCUSSION

#### 1. Absorption spectra

Figure 1 shows the absorption spectra obtained for 50 ns and 1  $\mu$ s after the pulse for LiCl-KCl eutectic

melt at 400°C. In the spectrum obtained for 50 ns after the pulse, two absorption bands appeared in u.v. and in the visible region. But at 1  $\mu$ s after the pulse, the visible band had diminished and only the u.v. band was observed. The difference spectrum in Fig. 1 is assigned to the visible band. These results revealed that the two kinds of transient species having different life times existed in the irradiated system. Furthermore, it was found that the bands were distinguishable spectrally.

#### 2. Kinetic analysis

The kinetic trace obtained at 700 nm is shown in Fig. 2. According to this trace, it was found that the species around 700 nm were produced during the pulse and the life time was about 250 ns. The decay was well fitted to the first order kinetic, and the reaction rate constant was estimated to be  $k_1 = 4.5 \times 10^6 \text{ s}^{-1}$ .

In Fig. 3, the kinetic trace obtained at 340 nm is shown as curve (a). The initial part of this trace includes a certain amount of the contribution of the species responsible for the visible band. Curve (b) shows the trace, which is devoid of the contribution found in trace (a). From curve (b), it was found that the species responsible for the u.v. band was formed after the pulse. Curve (c) is the simulated trace, assuming that this species was produced during the pulse and decayed by the second order kinetic. Curve (c) was observed to fit curve (b) well in the decaying part. Therefore, it was judged that the species decayed by the second order kinetic, and the reaction rate constant was estimated to be  $k_2/\epsilon \cdot l = 3.0 \times 10^6 \text{ s}^{-1}$ .

Further, the first order plots corresponding to the rising part in curve (b) were arranged well on the straight line. This result suggested that the species was formed by the first order kinetic, and the reaction rate constant was estimated to be  $k_1 = 2.5 \times 10^5 \text{ s}^{-1}$ .

This paper was presented at the 2nd Japan-China Bilateral Symposium on Radiation Chemistry, Osaka, Japan (September 1985).

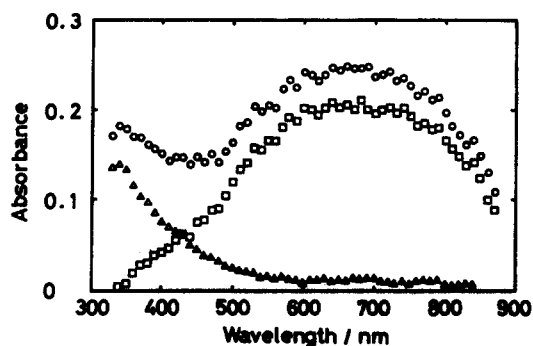


Fig. 1. Absorption spectra in LiCl-KCl eutectic (42 mol% KCl) melt at 400°C. ○, 50 ns after the pulse; △, 1 μs after the pulse; □, difference spectrum between 50 and 500 ns.

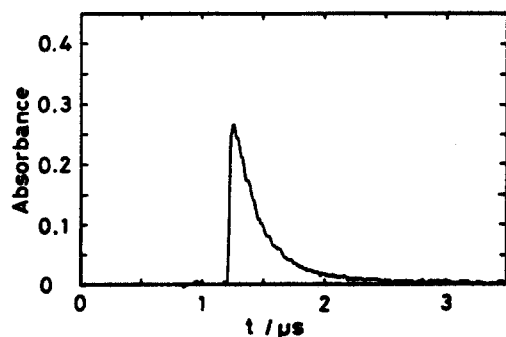


Fig. 2. Kinetic trace obtained at 700 nm in LiCl-KCl eutectic melt at 400°C.

### 3. Temperature dependence on the spectra

Figure 4 shows the temperature dependence on the visible band observed for 50 ns after the pulse using the LiCl-KCl eutectic mixture. At 400°C the peak position of the band appeared at 680 nm, and it shifted toward the red as the temperature increased. On the other hand, no temperature dependence on the u.v. band was observed in this temperature range. The peak position of the u.v. band was always observed at 340 nm.

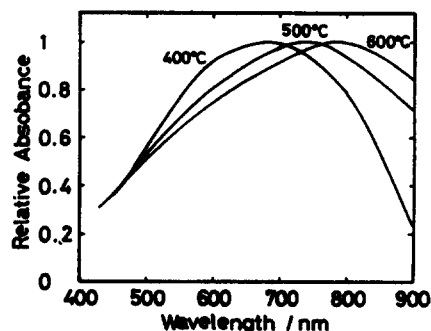


Fig. 4. The visible bands in LiCl-KCl eutectic melts at different temperatures.

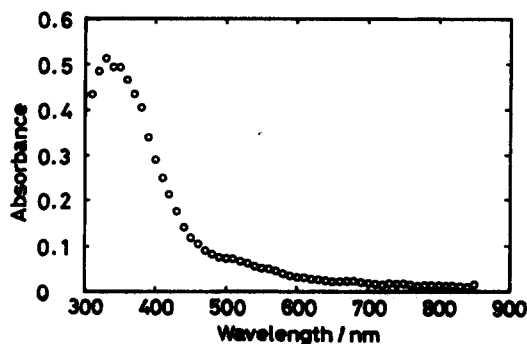


Fig. 5. Absorption spectrum obtained for 50 ns after the pulse in LiCl-KCl eutectic melt doped with 0.05 M CdCl<sub>2</sub> at 600°C.

Increasing temperature led to a decrease in density, or in other words, to an increase in specific volume. This result indicated that the species responsible for the visible band was affected by the change in the specific volume, but the species responsible for the u.v. band was not.

### 4. Effect on the spectra by doping with Cd<sup>2+</sup>

Figure 5 shows the spectrum obtained for 50 ns after the pulse in LiCl-KCl eutectic melt which was doped with 0.05 M CdCl<sub>2</sub> at 600°C. In this case, the

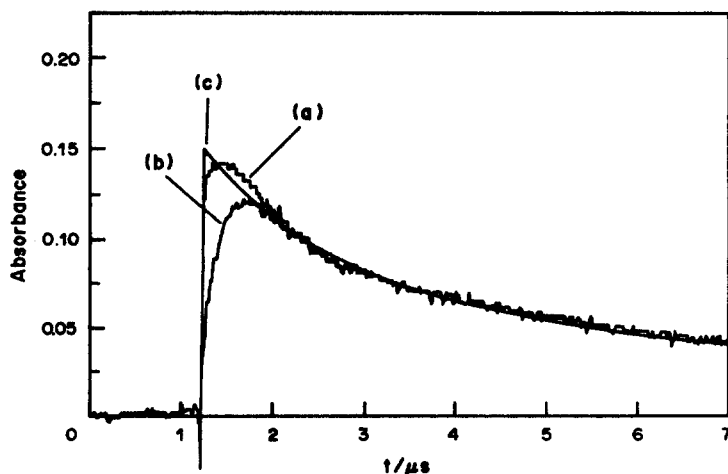


Fig. 3. (a) Kinetic trace obtained at 340 nm in LiCl-KCl eutectic melt at 400°C. (b) Kinetic trace which is devoid of the contribution as found in curve (a). (c) Simulated trace assuming the second order decay.

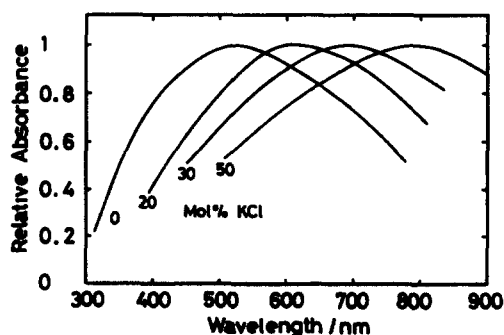


Fig. 6. The visible bands for several compositions of LiCl-KCl melt at 600°C.

visible band was not observed at all, and only the u.v. band, which was slightly blue, shifted, and was observed around 330 nm.  $\text{Cd}^{2+}$  is well known as a typical electron scavenger.<sup>(3)</sup> Therefore, it is reasonable to consider that the species responsible for the visible band is a kind of reducing species.

##### 5. Composition dependence on the spectra

In Fig. 6, the visible band obtained for several compositions of LiCl-KCl melts at 600°C are shown. In pure LiCl melt, the peak position of the visible band was observed around 520 nm. But it shifted to a longer wavelength in the LiCl-KCl mixture, and in the mixture whose composition of KCl was equal to 50 mol%, it appeared around 780 nm. Figure 7 shows the relation between the composition and the peak position of the visible band. The figure shows that the peak position shifted linearly toward the red corresponding to the mole fraction of KCl. However, as compared with the visible band, no shift of the u.v. band was observed in these experiments.

##### 6. Attribution of the spectra

As mentioned above, the species responsible for the visible band was formed during the pulse and decayed

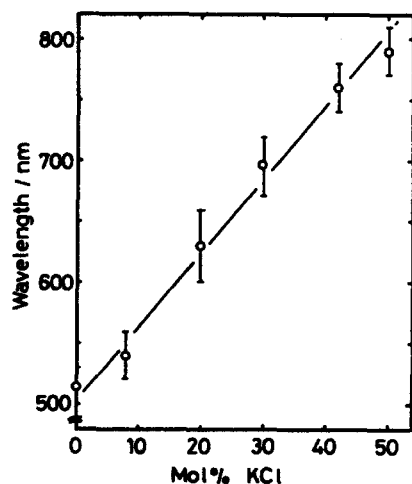


Fig. 7. The relation between the composition and the peak position of the visible bands.



Fig. 8. Arrhenius plots for the observed rate constants of  $e_s^-$  decay.

by the first order kinetic. In addition, the band was affected by the change in the specific volume and composition. Further, the band was diminished in the presence of  $\text{Cd}^{2+}$ , which is an electron scavenger. These facts and the experimental results obtained by Pikaev *et al.* using other alkali halide melts<sup>(1)</sup> suggested that the species responsible for the visible band was the solvated electron  $e_s^-$ . The  $e_s^-$  is considered to be an electron trapped by several cations.

On the other hand, the species responsible for the u.v. band was formed by the first order kinetic after the pulse, and decayed by the second order kinetic. Also the effect of temperature and composition on the band shift was not observed. These facts and the experimental results in NaCl aqueous solution<sup>(4)</sup> indicated that the species responsible for the u.v. band was considered to be the molecular radical ion  $\text{Cl}_2^-$ .

##### 7. Reaction mechanism

Figure 8 shows the Arrhenius plots for the observed rate constants of the solvated electron decay in LiCl-KCl eutectic melt. At temperatures below 500°C, the plots gave a straight line, and the activation energy of the reaction was estimated to be  $E_A = 43$  kJ/mol. But above 500°C, the plots showed a curve. This means that the activation energy apparently decreased above 500°C. This fact is explained by the existence of the reverse reaction, which occurs more easily as the temperature increases.

Figure 9 shows the Arrhenius plots for the decaying rate constants of  $\text{Cl}_2^-$ . In the pure melt, a straight line appears in the temperature range of this experiment. Here the activation energy was estimated to be  $E_A = 24$  kJ/mol. But in the melt containing  $\text{Cd}^{2+}$ , the plots did not show a straight line below 500°C. These facts indicate that the  $\text{Cl}_2^-$  decayed corresponding to the different reaction in the doped melt from the one in the pure melt.

The reaction schemes in radiolysis of alkali halide melts were proposed by Pikaev *et al.*<sup>(1)</sup> as follows:



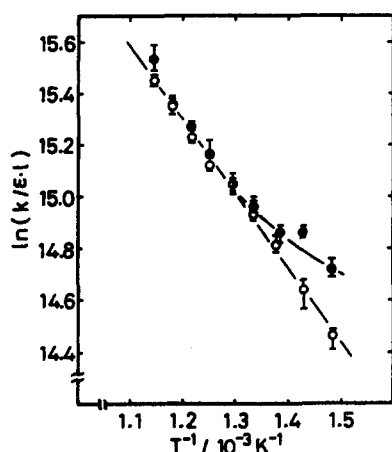
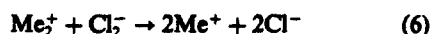
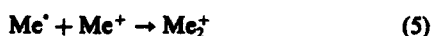
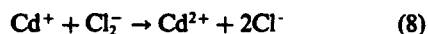


Fig. 9. Arrhenius plots for the observed rate constants of  $\text{Cl}_2^-$  decay. O, in pure LiCl-KCl eutectic melt; ●, in LiCl-KCl eutectic melt doped with 0.05 M  $\text{CdCl}_2$ .



where, Me is the alkali metal cation in the system. According to our results, the  $\text{e}_i^-$  was formed during the pulse and reduced by the first order kinetic. And the  $\text{Cl}_2^-$  was formed by the first order kinetic and reduced by the second order kinetic. Therefore, these schemes were confirmed by our results.

Then in the melt doped with  $\text{Cd}^{2+}$ , the following schemes were suggested:<sup>(1)</sup>



Reaction (7) indicates the electron scavenging reaction by  $\text{Cd}^{2+}$  and the formation of  $\text{Cd}^+$ . It has been reported that  $\text{Cd}^+$  has an absorption band around 300 nm in water at room temperature.<sup>(5)</sup> The spectra obtained in the eutectic melt doped with  $\text{Cd}^{2+}$  shows the absorption band around 330 nm (see Fig. 5). From these facts, the 330 nm band appears to be contributed by  $\text{Cd}^+$  as well as by  $\text{Cl}_2^-$ , which is around 340 nm.

Finally, the possibility of chlorine gas generation was considered as follows:



An experiment in which the LiCl-KCl eutectic melt at 400°C was continuously irradiated by about 10,000 pulses was carried out. It was found that a certain amount of gas was generated in the system, but the yield of the gas not measured quantitatively. This problem will have to be solved in the near future.

*Acknowledgements*—The authors wish to thank Dr Takeaki Enoto and Messrs Hiroaki Tanida, Koichi Sato and Masatoshi Kitaichi for their help in operating the linac.

#### REFERENCES

1. A. K. Pikaev, I. E. Makarov and T. N. Zhukova, *Radiat. Phys. Chem.* 1982, 19, 377.
2. E. Elchardus and P. Laffitte, *Bull. Soc. Chim. France* 1932, 51, 1572.
3. R. K. Wolff, M. J. Bronskill and J. W. Hunt, *J. Chem. Phys.* 1970, 53, 4211.
4. L. I. Grossweiner and M. S. Matheson, *J. Phys. Chem.* 1957, 61, 1089.
5. R. K. Wolff, J. E. Aidrich, T. Penner and J. W. Hunt, *J. Phys. Chem.* 1975, 79, 210.